

Using DB Fiddle for Your SQL Assignments

If you are having trouble installing Docker or PostgreSQL locally, you can still complete your SQL work using **DB Fiddle**—an online SQL playground server that supports PostgreSQL.

Step 1: Go to <https://www.db-fiddle.com/>

Step 2: Select the Proper Database

In the dropdown on the top-left, select: PostgreSQL 17 (or the latest available PostgreSQL version).

Database: PostgreSQL v17

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5.5
PostgreSQL
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Fiddle Title: PostgreSQL
50 characters remaining.
Fiddle Description: 300 characters remaining.
Private Fiddle PRO
This setting cannot be modified after saving the fiddle.
Upgrade to PRO
50% OFF for Early Adopters
Show Keyboard Shortcuts

133 INSERT INTO WORKS_ON (ESSN, PNO, HOURS) VALUES
134 ('123456789', 1, 10),
135 ('234567891', 2, 20),
136 ('345678912', 3, 15),
137 ('456789123', 4, 18),
138 ('567891234', 5, 12),
139 ('678912345', 6, 25),
140 ('789123456', 7, 30),
141 ('891234567', 8, 14),
142 ('912345678', 9, 16),
143 ('999999999', 10, 22);

140 INSERT INTO DEPENDENT (ESSN, DEPENDENT_NAME, SEX, BDATE, RELATIONSHIP) VALUES
141 ('123456789', 'Amy', 'F', '2010-01-01', 'Daughter'),
142 ('123456789', 'Tom', 'M', '2008-05-23', 'Son'),
143 ('234567891', 'Lucy', 'F', '2012-03-14', 'Daughter'),
144 ('345678912', 'Mike', 'M', '2011-07-07', 'Son'),
145 ('456789123', 'Emma', 'F', '2009-09-09', 'Spouse'),
146 ('567891234', 'Nina', 'F', '2013-12-25', 'Daughter'),
147 ('678912345', 'Josh', 'M', '2007-06-30', 'Son'),
148 ('789123456', 'Eli', 'M', '2015-11-11', 'Son'),
149 ('891234567', 'Sophie', 'F', '2016-04-04', 'Daughter'),
150 ('999999999', 'Henry', 'M', '2014-08-08', 'Spouse');

Query SQL
1 select *
2 from employee;

Text to DDL

Results
Query #1 Execution time: 0.23ms

fname	minit	lname	ssn	bdate	address
John	B	Smith	123456789	1980-01-15	123 Main St
Anna	L	Brown	234567891	1979-02-20	456 Oak Ave

Step 3: Set Up the Schema and Data for Problem #1

Copy and paste the schema and sample data provided to you in the *init.sql* file into the *Schema SQL* pane. Then click “Run” to create the tables and insert data.

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Schema SQL

```
1 -- Create tables
2 DROP TABLE IF EXISTS EMPLOYEE CASCADE;
3 CREATE TABLE EMPLOYEE (
4     FNAME VARCHAR(20),
5     MINIT CHAR(1),
6     LNAME VARCHAR(20),
7     SSN CHAR(9) PRIMARY KEY,
8     BDATE DATE,
9     ADDRESS VARCHAR(100),
10    SEX CHAR(1),
11    SALARY INTEGER,
12    SUPER_SSN CHAR(9) REFERENCES EMPLOYEE(SSN)
13        ON UPDATE CASCADE
14        ON DELETE SET NULL,
15    DNO INT
16 );
17
18 DROP TABLE IF EXISTS DEPARTMENT CASCADE;
19 CREATE TABLE DEPARTMENT (
20     DNAME VARCHAR(30),
21     DNUMBER INT PRIMARY KEY,
22     MGR_SSN CHAR(9) REFERENCES EMPLOYEE(SSN)
23         ON UPDATE CASCADE
24         ON DELETE SET NULL,
```

Text to DDL

- You only need to do this once per session. Once built, your tables are ready for querying.
- **Verify Your Tables**
 - To confirm that the tables were created successfully, try running a simple query like `'SELECT * FROM table_name;'` in the *Query SQL* pane.
 - Replace *table_name* with the name of one of your tables. This should return a few rows from that table if everything is set up correctly.

Step 4: Write Your Queries for Problem #1

Use the *Query SQL* pane in the right to write your SQL queries. Click “Run” to see the results of your queries to be displayed in the *Results* pane.

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Schema SQL

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20     DNAME VARCHAR(30),
21     DNUMBER INT PRIMARY KEY,
22     MGR_SSN CHAR(9) REFERENCES EMPLOYEE(SSN)
23     ON UPDATE CASCADE
24     ON DELETE SET NULL,
```

Text to DDL

Query SQL

```
1 SELECT
2 FROM EMPLOYEE
3 WHERE SALARY > 60000;
```

Results

Query #1 Execution time: 0.31ms

fname	minit	lname	ssn	bdate	address	sex	salary	super_ssn	dno
Anna	L	Brown	234567891	1979-02-20	456 Oak Ave	F	62000	123456789	2
Susan	T	Young	678912345	1975-06-12	987 Spruce Dr	F	73000	null	3

Copy as Markdown

- Make sure to save your work frequently!

Step 5: Create Your Own Database Schema and Data for Problem #2